Physical Chemistry CUME
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Assorted thermodynamic questions

A. A waterfall has an 85 m drop. Essentially all the energy of the fall goes into warming the water. How much warmer is the tailwater than the headwater, in °C? The specific (per kg) heat capacity of water is C_p = 4.184 kJ/(K kg), the gravitational acceleration is g = 9.81 m s^{-2}. (20 p)

B. The standard molar enthalpy of formation of common salt (塩 in Chinese), NaCl (s) is -411.15 kJ/mol at 298 K. The constant pressure molar heat capacity of Na(s) is $C_{p,m}$ = 28.24 J K^{-1} mol^{-1} **(note: J, not kJ),** that of Cl_2 is 33.91 J K^{-1} mol^{-1}, and that of NaCl is 50.50 J K^{-1} mol^{-1}. What is the standard enthalpy of formation of NaCl at 600 K? (20 p)
(Note: neither NaCl nor Na have molecules in the solid phase. 1 mole is customarily taken as the amount corresponding to the formula mass, e.g., 23 g Na, the same as the atomic mass of Na).

C. The vapor density of melting sulfur at 115.35 °C is very low but measurable, 2.10×10^{-24} (2.10E-24) g/L, and its vapor pressure is 2.65×10^{-20} Pa (2.65E-20 Pa). What is the molecular formula of sulfur in the gas phase at 115 °C? (E.g., S, S_2, S_3, ?) R=8.3145 J K^{-1} mol^{-1}. The atomic mass of sulfur is 32.07 g/mol. (30 p)

D. A heat pump is a heat engine running in reverse, moving heat from a colder place to a warmer one, and consuming electrical or mechanical energy. If the room temperature is +24 °C, and the outside temperature is -10 °C, how much more efficient is a heat pump than simple electrical heating, assuming 100 % efficiency, i.e. reversibility? The efficiency is the amount of heat energy delivered in the room divided by the amount of work consumed by the heat pump. Give it both as the ratio heat delivered/work invested, and as a per cent value (100% = 1) (30 p)